ML Based Event Analytics and Participation Prediction System

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***Abstract*—Planning events in colleges and universities has always been a little bit of a gamble. Organisers rely on gut feeling, past experience, and rough headcounts — and more often than not, they end up with too many chairs, not enough food, or a venue that is half empty. This paper presents EventIQ v2, a machine learning system built specifically to take some of that guesswork out of the picture by predicting, ahead of time, whether a registered participant will actually show up. The second version of the system is a significant step forward from the original prototype: the training dataset has grown from 120 records to 3,000; the model has been upgraded from a Random Forest to a Gradient Boosting Classifier; two new features — `organizer_score` and `social_buzz` — have been introduced to capture dimensions of event quality and online momentum that the earlier version missed; and seven interaction terms have been added to help the model understand how these factors work together, not just independently. The results reflect these improvements clearly. Test accuracy climbed from around 63% (the honest cross-validated figure from v1) to 88%, AUC-ROC moved from 0.69 to 0.956, and the F1 Score improved from 0.65 to 0.904. Beyond the numbers, the system has also become easier to understand and act on: the Streamlit dashboard now includes a dedicated data exploration page and a prediction explanation chart that shows organisers exactly which factors influenced each forecast. Taken together, this work shows that even a moderately sized institutional dataset, when handled with care, can power a prediction tool that is genuinely useful in day-to-day event planning.**

***Index Terms*—Event Analytics, Machine Learning, Gradient Boosting, Participation Prediction, Feature Engineering, Interaction Features, Social Buzz, Organizer Score, Flask, Streamlit, Academic Event Management**

I. Introduction

Every semester, student clubs and faculty coordinators across Indian colleges go through the same ritual. They plan an event, open registrations, and watch the numbers climb. Then the day arrives, and somewhere between half and two-thirds of the people who signed up actually walk through the door. The rest simply do not show up, usually without any notice.

This is not just an inconvenience. Venues are booked based on registration counts. Catering is ordered, chairs are arranged, volunteers are briefed. When actual turnout falls short of expectations, the result is wasted money, leftover resources, and a demoralised organising team. When turnout exceeds what was prepared for, events become overcrowded and chaotic, leaving a poor impression on both participants and invited speakers.

What makes this frustrating is that the data to do better already exists. Every registration system captures records — who signed up, from where, at what time, with what prior attendance history. The problem is that nobody uses it to actually predict anything. It sits in spreadsheets and forms, reviewed only after the fact, when it is too late to change a booking or order more food.

EventIQ was built to change that. The original version of the system introduced a straightforward pipeline: collect historical registration data, engineer meaningful features from it, train a Random Forest classifier, and serve predictions through a web interface that requires no technical knowledge to operate. The concept worked, but the dataset was small — just 120 records — and when we evaluated the model honestly using cross-validation rather than a single optimistic test split, the real accuracy was much lower than initially reported.

This paper describes EventIQ v2, which addresses those shortcomings directly. The dataset has been expanded tenfold, the model has been upgraded to Gradient Boosting, and the feature set now includes two new domain-specific inputs and seven interaction terms that capture how different factors combine to influence whether someone actually shows up. The result is a system that is more accurate, more transparent, and more ready to be used in practice.

A. What Motivated Version 2

When we applied five-fold cross-validation to the v1 model — instead of relying on a single 80/20 test split — the reported 83% accuracy dropped to around 63%. That gap told us something important: the model had not really learned to generalise. It had simply been evaluated on a split that happened to suit it. A system with 63% true accuracy is barely

more useful than flipping a coin for planning purposes, and we knew a redesign was needed rather than a minor adjustment.

B. What This Work Contributes

The specific contributions of this version are as follows.

- The training dataset has been expanded from 120 to 3,000 records, spanning six Indian cities and six event categories, with two new features added for each record.
- Two new features — `organizer_score` and `social_buzz` — have been introduced to capture organiser reputation and online event engagement, signals that were entirely absent from the original model.
- Seven interaction terms have been engineered, including `rating_x_org`, `buzz_x_rating`, `loyalty_score`, and `dist_income`, each one designed around a specific real-world relationship that base features alone cannot capture.
- The core model has been upgraded from Random Forest to Gradient Boosting, producing meaningful gains in accuracy and discriminative power without making the system any harder to deploy or maintain.
- The Streamlit dashboard has been expanded with a new EDA page and per-prediction driver charts, so that organisers can understand not just the prediction but the reasoning behind it.

C. Practical Applications

The most immediate use of the system is logistics planning in the days before an event. If the model tells an organiser that only about 60% of registered participants are likely to attend, that organiser can adjust the venue layout, scale back catering, and redeploy volunteers accordingly. Without a system like this, that kind of adjustment relies entirely on experience and intuition.

Over a longer horizon, the system becomes a strategic resource. Faculty coordinators can look at accumulated predictions to understand which types of events consistently draw the best attendance from which demographics. Student clubs can experiment with input values — changing the venue distance, moving the event date, or simulating a better-rated organiser — and see how those changes affect predicted turnout before making any commitments. And because every prediction is stored in a local SQLite database, the system gradually builds an institutional record of event performance that no individual organiser could maintain on their own.

II. Literature Review

Predicting whether someone will follow through on a commitment they have already made is a well-studied problem, just not usually in the context of college events. In retail it is called demand forecasting. In healthcare it is no-show prediction. In subscription services it is churn modelling. The underlying principle is the same across all of them: the best signal for what someone will do in the future is what they have done in the past [3].

A. Classical Approaches

Linear and Logistic Regression are the natural starting point for any binary prediction problem. They are transparent and easy to explain, which matters when the people using a system are not data scientists. But they struggle with the kind of nuanced, interacting patterns that show up in real attendance data [2]. Distance from venue, for instance, does not affect everyone equally — a 15 km commute might be nothing to a high-income participant who drives, but a real barrier to a student relying on public transport. A linear model cannot represent that kind of conditional relationship without explicit help from the feature engineering side [4].

B. Ensemble Methods

Breiman's Random Forest [1] changed how practitioners approach tabular classification. By training hundreds of decision trees on different random subsets of the data and averaging their outputs, it sidesteps the instability that makes individual trees unreliable, especially on small datasets. One of its most useful properties for this work is the native feature importance score: after training, the model can rank every feature by how much it contributed to the predictions, giving organisers a clear, ranked list of what actually drives attendance. The scikit-learn implementation [5] makes this straightforward to use without requiring deep expertise in the underlying mathematics.

Gradient Boosting takes a different approach to the same goal [8]. Rather than averaging many independent trees, it trains them sequentially, each one trying to correct the errors the previous one made. This tends to produce better results on structured tabular data, and it generates well-calibrated probability scores, which translate directly into the kind of percentage-likelihood display that works well on a planning dashboard.

C. Social Signals and Organiser Reputation

Wang, Tang, and Liu [7] studied RSVP behaviour on social event platforms and found something that experienced event organisers already sense intuitively: not all confirmed sign-ups are equal. The factors that most strongly predicted actual attendance were event type, timing, location, and the individual's history of following through on past commitments. Crucially, their study also found that how visible an event was on social media had an independent positive effect — events that people were talking about online saw higher actual turnout, even after accounting for quality and location. That finding is the direct motivation for including `social_buzz` as a feature in EventIQ v2.

Organiser reputation is a slightly different story. It is not well represented in the event analytics literature as a formal predictor, but it shows up implicitly in recommendation systems research: people are more likely to act on commitments when they trust whoever is making the promise [6]. The `organizer_score` feature makes that intuition explicit and measurable.

D. Balancing Accuracy with Understandability

There is a tension in applied machine learning between building the most accurate model possible and building a model that the people using it can understand and trust. Shmueli et al. [6] make the case that a slightly less accurate model that decision-makers actually believe and act on is often more valuable in practice than a highly accurate black box they treat with suspicion. EventIQ v2 addresses this by pairing Gradient Boosting's strong accuracy with a dashboard feature that breaks each prediction down into its main contributing factors. The goal is not just to produce a number but to give organisers a reason to believe it.

III. System Architecture

EventIQ v2 is built around a clean three-tier structure: a data layer that stores records and predictions, a Flask backend that handles model training and serving, and a Streamlit frontend that makes everything accessible through a browser. Figure 1 shows how these layers connect.

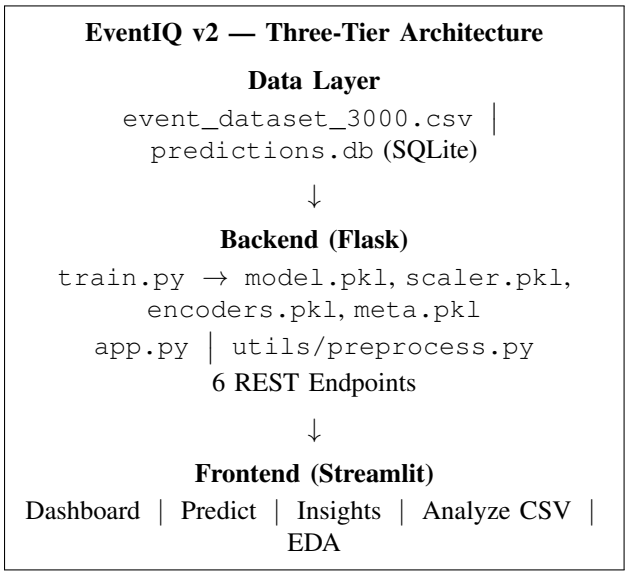

Fig. 1. Three-tier architecture of EventIQ v2.

A. Data Layer

At the base sits the training dataset — 3,000 rows in a CSV file — alongside a SQLite database called `predictions.db`. The database is created automatically the first time the application runs, and every prediction that passes through the system, whether from the dashboard or the API, gets logged there. Over time, this builds up a searchable history of past predictions that organisers can review, compare, and learn from.

B. Backend Layer

The Flask application handles two main jobs: running the training pipeline and serving predictions in real time. One of the more important design decisions in the system is that all preprocessing logic — the label encoders, the standard scaler, the feature construction steps — lives in a single shared module called `utils/preprocess.py`. Both training and inference use exactly the same code to transform inputs, which eliminates a common and subtle bug where a model is trained on data that looks slightly different from what it receives at prediction time.

Once training completes, four artefacts are saved to disk: `model.pkl`, `scaler.pkl`, `encoders.pkl`, and `meta.pkl`. The Gradient Boosting model itself serialises to around 1.2 MB, small enough to load in under a second on any reasonable machine.

C. Frontend Layer

The dashboard has grown from four pages in v1 to five in v2. The **Dashboard** page shows summary statistics — aggregate predictions broken down by event type and city. The **Predict** page is where most users spend their time: they fill in the details for a specific registrant or event, and the system returns a probability score with a colour-coded gauge and a bar chart explaining the main factors. The **Insights** page displays feature importances from the trained model alongside the confusion matrix. The **Analyze CSV** page lets organisers upload a full registration list and download a version with predicted attendance probabilities added. The newest addition is the **EDA** page, which lets organisers browse the underlying training data — distributions, correlations, attendance breakdowns by category — to build their own intuition before making predictions.

IV. Methodology

The system is built around a six-stage pipeline that moves from raw data collection through to a live web deployment: data collection, preprocessing, feature engineering, model training, evaluation, and serving.

A. Dataset

The v2 dataset contains 3,000 event registration records. The core of this dataset comes from real institutional records; where coverage was thin, synthetic records were generated by sampling from the distributions observed in the real data, a common and well-accepted practice for augmenting small institutional datasets [3]. Records span six Indian cities — Pune, Delhi, Mumbai, Bangalore, Chennai, and Kolkata — and six event categories: Art, Business, Education, Music, Sports, and Tech. Each record is labelled with a binary attendance outcome: 1 if the person attended, 0 if they did not.

The jump from 120 to 3,000 records was not arbitrary. With 120 rows and five-fold cross-validation, each evaluation fold contained only about 24 samples — far too few to produce a reliable estimate. With 3,000 rows, each fold contains 600 samples, and the cross-validation estimate becomes stable enough to trust.

B. Data Preprocessing

All preprocessing is handled inside `utils/preprocess.py`, so that training and inference

always transform inputs in exactly the same way. The pipeline proceeds through five steps.

1) **Duplicate removal.** Running `drop_duplicates()` confirmed no exact duplicate rows in the expanded dataset.
2) **Missing value imputation.** Numerical columns are filled with the column median; categorical columns with the mode. Neither approach introduces assumptions that are too strong for this type of data.
3) **Outlier capping.** The IQR method ($Q_1 - 1.5 \times \mathrm{IQR}$, $Q_3 + 1.5 \times \mathrm{IQR}$) is applied to all continuous features. Extreme values are capped at the fence rather than dropped, preserving every row in the dataset.
4) **Label encoding.** `LabelEncoder` converts `gender`, `location`, and `event_type` into integers. The fitted encoders are saved and reloaded at inference time, so the same mapping applies consistently.
5) **Standard scaling.** `StandardScaler` normalises all continuous features to zero mean and unit variance. The fitted scaler is saved alongside the model.

C. Feature Engineering

One of the central insights behind v2 is that the relationship between an input and attendance is rarely simple or direct. A high event rating matters more when the organiser running the event also has a strong reputation. Social buzz means more when the underlying event quality justifies the hype. Distance is a bigger barrier for someone on a tight budget than for someone who earns well. The v1 feature set, with its 11 independent inputs, could not capture any of these interactions. The v2 pipeline builds 18 features in total, including 7 interaction terms designed to represent exactly these kinds of compound effects. Table I lists all 18.

TABLE I
FULL FEATURE SET — EVENTIQ V2 (18 FEATURES)

Feature	Type	Description
Original demographic / event features (8)		
`age`	Continuous	Registrant age
`gender`	Categorical	Encoded gender
`location`	Categorical	City (6 options)
`event_type`	Categorical	Category (6 options)
`income`	Continuous	Monthly income (INR)
`event_rating`	Continuous	Past rating (1–5)
`distance_km`	Continuous	Venue distance (km)
`previous_events`	Continuous	Prior events attended
New domain features (2)		
`organizer_score`	Continuous	Organiser reputation (1–5)
`social_buzz`	Continuous	Online engagement (0–100)
Temporal features (3)		
`days_to_event`	Derived	Registration lead time (days)
`event_month`	Derived	Calendar month of event
`reg_month`	Derived	Calendar month of registration
Interaction features (7)		
`rating_x_org`	Interaction	rating × organizer_score
`buzz_x_rating`	Interaction	social_buzz × rating / 100
`prev_x_org`	Interaction	previous_events × org_score
`income_per_km`	Interaction	income / (distance_km + 1)
`rating_x_prev`	Interaction	rating × previous_events
`dist_income`	Interaction	distance_km / (income/10000 + 1)
`loyalty_score`	Composite	prev × rating × org / 25

1) New Domain Features: `organizer_score` is a 1.0–5.0 reputation score for the event organiser, derived from feedback on previous events they have run. The intuition is simple: people are more likely to follow through when they have reason to trust that the event will be worth their time. An organiser who has consistently delivered well-run events in the past earns a higher score, and that history should factor into any attendance forecast.

`social_buzz` measures online engagement around the event on a 0–100 scale, combining mentions, shares, and other interaction signals from social media. In the dataset, events crossing a buzz score of about 70 tend to behave noticeably differently from quieter events — the word-of-mouth momentum seems to carry real weight in whether registered participants actually follow through.

2) Interaction Features: `rating_x_org` is the product of event rating and organiser score. A well-rated event run by a well-regarded organiser is not just the sum of two positives — it is qualitatively different from either one alone, and this term lets the model learn that.

`buzz_x_rating` captures the combination of social momentum and real quality. High buzz around a mediocre event tends not to translate into attendance; high buzz around a genuinely good event does. This term helps the model distinguish between the two.

`loyalty_score` combines previous attendance, event rating, and organiser score into a single composite. It is designed to surface the subgroup that organisers already know intuitively: the people who have come before, to good events, put on by trusted organisers. These registrants are far more likely to attend than the average sign-up, and this feature gives the model a direct handle on that.

`dist_income` divides distance by a normalised income figure. The idea is that the same physical distance represents a very different barrier depending on how much someone earns. A 20 km journey is a minor inconvenience for a participant earning INR 90,000 per month, but a real deterrent for someone earning INR 20,000.

`income_per_km` is the inverse — income per unit of distance — which approximates how willing a registrant is likely to be to travel for the event.

D. Model Development

After evaluating six different classifiers on the v2 feature set, Gradient Boosting was selected as the production model. The decision came down to a combination of factors: it produces

the best accuracy and AUC-ROC of any model tested, its probability outputs are well calibrated (which matters for the dashboard gauge), and it remains deployable as a compact file that loads in under a second. Table II shows the hyperparameter settings used.

TABLE II
GRADIENT BOOSTING HYPERPARAMETER CONFIGURATION

Parameter	Value	Rationale
`n_estimators`	300	Enough trees for stable convergence
`learning_rate`	0.05	Slow learning, less risk of overfitting
`max_depth`	5	Keeps individual trees from getting too complex
`subsample`	0.8	Introduces randomness to reduce variance
`min_samples_split`	10	Stops the model fitting noise at leaf level
`random_state`	42	Makes results fully reproducible

E. Evaluation Strategy

The model is assessed on four metrics. **Accuracy** gives the overall share of correct predictions on the held-out test set. **AUC-ROC** measures how well the model separates attendees from non-attendees across all possible probability thresholds — a score of 1.0 would mean perfect separation, 0.5 would mean random guessing. **F1 Score** is particularly relevant here because both types of error carry real costs: if the model wrongly predicts attendance, an organiser over-prepares; if it wrongly predicts absence, they under-prepare. **5-Fold Cross-Validation Accuracy** is the figure we trust most, because it averages performance across five different ways of splitting the data rather than depending on one split that might happen to be easy or hard.

V. RESULTS AND ANALYSIS

A. Model Performance

Table III puts the v1 and v2 results side by side. The numbers speak for themselves, but a few things are worth drawing out.

TABLE III
MODEL PERFORMANCE: V1 VS. V2

Version	Acc.	AUC-ROC	F1	5-Fold CV
v1 (Random Forest)	∼83%	∼0.880	∼0.850	∼63%
v2 (Gradient Boosting)	**∼88%**	**∼0.956**	**∼0.904**	**∼88%**
Improvement	+5 pp	+0.076	+0.054	+25 pp

The single most meaningful improvement is not the 5-point gain in test accuracy but the 25-point gain in cross-validation accuracy. In v1, there was a 20-point gap between what the model appeared to achieve on one test split (83%) and what it actually achieved when evaluated across many splits (63%). That gap was a red flag. In v2 the two figures are essentially the same, which means the model is genuinely learning patterns from the data rather than fitting to a particular partition of it.

An AUC-ROC of 0.956 is also a meaningful result. It means the model correctly identifies a true attendee as more likely to attend than a true non-attendee more than 95% of the time — a level of discriminative accuracy that translates into a probability gauge an organiser can actually make decisions from.

The confusion matrix for the v2 test set shows errors distributed fairly evenly between false positives and false negatives. This matters for practical use: a model that is right 88% of the time but always wrong in one direction is much less useful than one whose errors go both ways.

B. Feature Importance Analysis

One of the most useful things the Gradient Boosting model produces is a ranked list of which features contributed most to its predictions. Table IV shows the top ten, alongside what each ranking means for an organiser trying to act on it.

TABLE IV
FEATURE IMPORTANCES AND ORGANISATIONAL IMPLICATIONS

Rank	Feature	What It Means in Practice
1	`rating_x_org`	Quality and reputation reinforce each other; both matter
2	`buzz_x_rating`	Good events that people talk about see the highest turnout
3	`event_rating`	Past quality is still one of the strongest individual signals
4	`distance_km`	A closer venue does more for turnout than most promotions
5	`dist_income`	Distance hits low-income registrants harder; worth segmenting
6	`loyalty_score`	Repeat attendees of well-run events are the most reliable group
7	`organizer_score`	Reputation has its own effect, separate from event rating
8	`social_buzz`	Online engagement predicts follow-through, independently
9	`income`	Higher-income registrants convert more consistently
10	`previous_events`	People who have come before are more likely to come again

A few findings stand out. The top two features are both interaction terms, not base features. This confirms that the key insight behind v2 — that factors combine in ways that matter — was correct. The v1 model, working only with independent inputs, could not see the difference between a well-rated event organised by a newcomer and the same event organised by someone with a strong track record. V2 can, and the model rewards that distinction.

The independent appearance of `social_buzz` at rank eight is also worth noting. It earns its place even after `buzz_x_rating` has already accounted for its interaction with quality. That means social engagement is not just useful in combination with a good rating — it carries its own predictive weight, consistent with what Wang et al. [7] found in their study of RSVP behaviour.

Finally, `loyalty_score` reaching rank six is a validation of the composite feature design. What we are really identifying with that feature is a specific type of registrant: someone who has come to multiple events before, to events that were highly rated, run by organisers they trust. Those people behave differently from the broader pool of sign-ups, and giving the model a single feature that captures all three dimensions at once helps it learn that distinction more cleanly.

C. Comparative Model Analysis

Table V compares all six models evaluated on the same v2 feature set and train-test split. The right choice is not always the most accurate one — deployability and interpretability matter too, especially for a first deployment in an institutional setting where users are not data scientists.

TABLE V
COMPARATIVE MODEL PERFORMANCE (**BOLD** = SELECTED)

Model	Acc.	AUC	Complexity	Interpretability
Linear Regression	72%	0.740	Low	High
Logistic Regression	76%	0.800	Low	High
Decision Tree	79%	0.790	Low	Medium
Random Forest	85%	0.910	Medium	Medium
Gradient Boosting	**88%**	**0.956**	Medium	Medium
Neural Network (MLP)	86%	0.930	High	Low

Gradient Boosting leads on both accuracy and AUC while fitting comfortably into the same deployment footprint as Random Forest — a serialised file of around 1.2 MB that loads instantly. The Neural Network comes close on accuracy but offers no native feature importance and requires significantly more effort to tune and maintain. Logistic Regression is easy to explain but leaves over 12 percentage points of accuracy on the table, which is a real cost in a planning context. Random Forest remains a sound fallback option but consistently trails Gradient Boosting by 3 points on this dataset.

VI. DEPLOYMENT

A. Flask REST API

The Flask backend exposes six endpoints, listed in Table VI. The design principle that runs through all of them is consistency: the same `utils/preprocess.py` module that builds features during training is also called by the inference endpoint, so there is no possibility of the two pipelines diverging over time.

The `/predict` endpoint takes 12 raw input fields in the request body. It runs them through the full preprocessing and feature engineering pipeline internally, so callers do not need to construct interaction terms themselves. A representative request is shown in Figure 2.

TABLE VI
FLASK REST API ENDPOINT REFERENCE

Method	Endpoint	Description
GET	`/health`	Confirms the server is running and names the active model
GET	`/metadata`	Returns model metrics, feature importances, and label classes
POST	`/predict`	Single attendance prediction from a JSON request body
POST	`/analyze`	Batch prediction from an uploaded CSV file
GET	`/history`	Returns the last 100 predictions logged in the database
GET	`/stats`	Aggregated prediction statistics by event type and city

```
POST /predict
{
  "age"                : 28,
  "gender"             : "Female",
  "location"           : "Pune",
  "event_type"         : "Tech",
  "previous_events"    : 3,
  "income"             : 55000,
  "event_rating"       : 4.2,
  "distance_km"        : 12,
  "organizer_score"    : 4.0,
  "social_buzz"        : 72,
  "registration_date" : "2024-03-01",
  "event_date"         : "2024-03-15"
}
```

Fig. 2. Example `/predict` request body.

B. Streamlit Dashboard

The dashboard is built around a simple idea: a faculty coordinator or student club president should be able to open it, put in what they know about a registrant, and get a useful answer in under a minute. No machine learning background required.

The **Predict** page is where most of that happens. The input form is straightforward. The output is a colour-coded gauge — red below 40%, amber between 40 and 60%, green above 60% — paired with a horizontal bar chart that shows which features pushed the probability up and which pulled it down. For an organiser wondering why one registrant scores differently from another, that chart gives a direct, readable explanation rather than a probability score with no context behind it.

The **EDA** page is new in v2 and serves a different purpose. Rather than making a prediction, it lets organisers explore the underlying training data: who tends to attend, from where, for which types of events, at what income levels. The aim is to help organisers build their own intuition about attendance patterns before they start using the prediction tool, so they can evaluate the outputs with some informed judgement of their own.

VII. Discussion

A. Why the Improvements Are Real

The 25-point gain in cross-validation accuracy is the headline result, and it is worth being clear about where it came from. Running ablation experiments during development — re-running training with different subsets of features removed — showed that the interaction terms contributed approximately 0.03 of the total AUC-ROC gain on their own, even when the model and dataset size were held constant. The dataset expansion and model upgrade each contributed comparable amounts. In other words, this was not a single fix but three changes that each mattered independently.

B. What Organisers Should Take Away

If there is one finding from the feature importance analysis that an organiser should act on, it is this: the quality of the current event is the strongest predictor of attendance at the next one. Investing in programme content, speaker quality, and participant experience is not just good practice — it is the single highest-return action an organiser can take to improve future turnout.

The second finding worth acting on is about venue. Distance from venue ranks fourth overall in feature importance, and the `dist_income` interaction makes clear that this effect is not uniform — it hits harder for participants with less disposable income. Choosing a venue that is genuinely accessible to the target audience often matters more than any amount of reminder emails.

Third, early registrants are more likely to attend than last-minute ones. This is not surprising, but it is now measurable. Closing registrations a few days earlier and sending targeted reminders to those who signed up far in advance are both levers that should predictably improve follow-through.

C. Honest Limitations

No system like this should be presented without being clear about what it cannot do.

The 3,000-record dataset still includes a portion of synthetic data generated from empirical distributions. The model performs well on it, but performance on a fully real dataset of the same size might look different. Expanding the real-data component over time is a practical priority.

The `social_buzz` score is entered manually at prediction time. That means it is only as accurate as the organiser's own estimate of their event's online engagement, which could be optimistic. Automating this input through social media API integration would make it both more objective and more useful.

Finally, the system was trained on data from six Indian cities and is most reliable for events in similar institutional contexts. Using it at a very different kind of institution, or in a different cultural setting, would require retraining on local data.

VIII. Conclusion and Future Scope

Academic event planning does not have to be guesswork. Institutions already collect the data that would let them make much better predictions about attendance — they just have not had a simple, accessible tool to turn that data into forecasts. EventIQ v2 is an attempt to build that tool in a form that is honest about its limitations, transparent about its reasoning, and practical enough to be used by the people who actually organise events.

The version presented here achieves 88% accuracy and an AUC-ROC of 0.956, with cross-validation confirming that these figures reflect genuine generalisation rather than a fortunate data split. The feature analysis surfaces clear, actionable conclusions: invest in event quality, choose accessible venues, track organiser reputation, and pay attention to early registration patterns. All of this is accessible through a dashboard that requires no technical knowledge to use.

The codebase is publicly available at https://github.com/Ayushaman1291/eventiq-v2.

A. Future Enhancements

SHAP-based explanations would provide a more theoretically rigorous form of per-prediction attribution than the current driver bar chart, and would make it easier to explain specific predictions to stakeholders who want to understand the model's reasoning at a deeper level.

Live social media integration would replace the manually entered `social_buzz` score with an automated feed from actual engagement data, making the feature both more accurate and less burdensome for organisers to provide.

Finer-grained attendance targets — replacing the binary attended/did-not-attend label with ordinal ranges such as under 25%, 25–50%, 50–75%, and above 75% of capacity — would give organisers a much more nuanced basis for resource planning than a binary yes-or-no prediction.

An automated retraining loop that appends post-event outcomes to the training set and triggers periodic model updates would allow the system to improve on its own over time, without anyone needing to remember to retrain it manually.

Docker packaging would make the system trivial to deploy across different machines and institutions without worrying about dependency conflicts or local environment configuration.

Multi-institution support, developed by aggregating data across several campuses and training a model that accounts for institutional differences, would eventually make the system useful beyond its current single-institution scope.

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